

Data Transformation, Manipulation and Preparation in SQL Cheatsheet

SELECT Choose which columns to return
Example `SELECT order_id, quantity FROM orders;`

FROM Specify the source table
Example `SELECT * FROM distributors;`

DISTINCT Return unique values only
Example `SELECT DISTINCT region FROM orders;`

AS Rename a column or expression
Example `SELECT COUNT(*) AS total_rows FROM orders;`

LIMIT Cap the number of rows returned
Example `SELECT * FROM orders LIMIT 10;`

Exploring & Profiling

-- Pattern
`SELECT COUNT(*) AS total_rows FROM your_table;`

-- How many rows do we have?
`SELECT COUNT(*) AS total_rows FROM orders;`

-- Pattern
`SELECT COUNT(*) AS total_rows, COUNT(column_name) AS rows_with_value FROM your_table;`

-- Spot the difference: COUNT(*) includes nulls, COUNT(col) doesn't
`SELECT COUNT(*) AS total_rows, COUNT(quantity) AS rows_with_quantity FROM orders;`

--Pattern
`SELECT LOWER(TRIM(column_name)), COUNT(*) FROM your_table GROUP BY 1HAVING COUNT(*) > 1;`

-- Find duplicates
`SELECT LOWER(TRIM(distributor_name)), COUNT(*)FROM distributors GROUP BY 1HAVING COUNT(*) > 1;`

REMEMBER:

COUNT(*) counts every row. COUNT(column) skips NULLs. This is one of the most common sources of wrong numbers in reporting.

ORDER BY Sort results — add DESC to reverse
Example `SELECT * FROM orders ORDER BY order_date DESC;`

Cleaning & Transforming

-- Pattern
`SELECT CAST (column AS datatype) AS alias FROM table;`

-- Convert text to number
`SELECT CAST(quantity_text AS INTEGER) AS quantity FROM orders;`

-- Pattern
`SELECT INITCAP(TRIM(table_name)) FROM table_name;`

-- How many rows do we have?
`SELECT INITCAP(TRIM(distributor_name)) FROM distributors;`

-- Pattern
`SELECT column_name, CASE WHEN column_name IS NULL THEN 'Missing' WHEN column_name > 2000 THEN 'Suspicious' WHEN column_name < 1 THEN 'Invalid' ELSE 'OK' END AS name_output_column FROM orders;`

-- Flag suspicious values with CASE
`SELECT order_id, CASE WHEN quantity IS NULL THEN 'Missing' WHEN quantity > 2000 THEN 'Suspicious' WHEN quantity < 1 THEN 'Invalid' ELSE 'OK' END AS quantity_flag FROM orders;`

--Pattern
`UPDATE table_name SET column_name = 'Tema Distributors' WHERE LOWER(TRIM(column_name)) IN ('tema dist.', 'tema distributors');`

--Safe update — always use WHERE
`UPDATE table_name SET column_name = 'Tema Distributors' WHERE LOWER(TRIM(column_name)) IN ('tema dist.', 'tema distributors');`

REMEMBER:

Never run UPDATE without a WHERE clause in production. Add a SELECT with the same WHERE first to preview what you'll change.

Joining Tables

-- Pattern
`SELECT a.col, b.col FROM table_a a INNER JOIN table_b b ON a.key = b.key;`

-- INNER JOIN - only matched rows
`SELECT d.distributor_name, o.order_id FROM distributors d INNER JOIN orders o ON d.distributor_id = o.distributor_id;`

-- Pattern
`SELECT a.col, COUNT(b.key) AS aliasFROM table_a a LEFT JOIN table_b b ON a.key = b.key GROUP BY a.col;`

-- LEFT JOIN: all distributors, even those with no orders
`SELECT d.distributor_name, COUNT(o.order_id) AS order_count FROM distributors d LEFT JOIN orders o ON d.distributor_id = o.distributor_id GROUP BY d.distributor_name;`

REMEMBER:

If your row count increases after a JOIN, you have a one-to-many relationship inflating results. Always check.

Aggregation & Dates

-- Pattern
`SELECT group_col, SUM(measure) AS total_alias, COUNT(*) AS count_alias FROM table GROUP BY group_col ORDER BY total_alias DESC;`

-- Group by with Sum & Count
`SELECT region, SUM(quantity) AS total_units, COUNT(*) AS order_count FROM orders GROUP BY region ORDER BY total_units DESC;`

-- Pattern
`SELECT group_col, SUM(measure) FROM table WHERE row_condition -- before grouping GROUP BY group_col HAVING SUM(measure) > n; -- after grouping`

-- Where vs Having
`SELECT region, SUM(quantity) FROM orders WHERE status <> 'Cancelled' GROUP BY region HAVING SUM(quantity) > 200;`

-- Pattern
`SELECT EXTRACT(part FROM date_col) AS alias FROM table;`

-- Extract Part of a Date
`SELECT EXTRACT(MONTH FROM order_date) AS order_month, EXTRACT(YEAR FROM order_date) AS order_year FROM orders;`

-- Pattern
`SELECT DATE_TRUNC('period', date_col) AS alias, SUM(measure) AS alias FROM table GROUP BY 1 ORDER BY 1;`

-- Extract Part of a Date
`SELECT DATE_TRUNC('month', order_date) AS order_month, SUM(quantity) AS total_units FROM orders GROUP BY 1 ORDER BY 1;`

-- Pattern
`SELECT group_col, CURRENT_DATE - MAX(date_col) AS alias FROM table GROUP BY group_col ORDER BY alias DESC NULLS LAST;`

-- Days Elapsed Since A Date
`SELECT distributor_name, CURRENT_DATE - MAX(order_date) AS days_since_last_order FROM orders o JOIN distributors d USING (distributor_id) GROUP BY distributor_name ORDER BY days_since_last_order DESC NULLS LAST;`

-- Pattern
`SELECT TO_CHAR(date_col, 'format') AS alias FROM table;`

-- Format a Date As Text
`SELECT TO_CHAR(order_date, 'YYYY-MM') AS month_label, TO_CHAR(order_date, 'Day DD Mon YYYY') AS full_label FROM orders;`

-- Pattern
`SELECT TO_CHAR(date_col, 'YYYY-MM') AS month, group_col, SUM(measure) AS alias FROM table GROUP BY 1, 2 ORDER BY 1, 2;`

-- Sales by month (time series)
`SELECT TO_CHAR(order_date, 'YYYY-MM') AS month, region, SUM(quantity) AS total_units FROM orders GROUP BY 1, 2 ORDER BY 1, 2;`